



WINFAB Diamondback 2018 is a three dimensional turf reinforcement mat (TRM).

WINFAB Diamondback 2018 is manufactured using a special heat stabilization process to drastically increase the effectiveness of the pyramidal weave when incorporated in steep slope, channelized flows, and vegetated applications where reinforcement is required for turf and erosion control.

WINFAB Diamondback 2018 is constructed using UV stabilized high tenacity monofilament yarns in a very consistent configuration which resists ultraviolet deterioration, rotting, biological degradation, and is inert to commonly encountered soil chemicals.

PROPERTY	TEST METHOD	ENGLISH	METRIC
Tensile Strength*	ASTM D6818	2,500 x 2,000 lb/ft	36.48 kN/m x 29.18 kN/m
Elongation	ASTM D6818	20 x 20%	20 x 20%
Thickness*	ASTM D6525	.25 in	6.4 mm
Resiliency	ASTM D6524	70%	70%
Light Penetration	ASTM D6567	30%	30%
Porosity	ECTC TASC 00197	96%	96%
UV Resistance (3,000 hrs)	ASTM D4355	90%	90%
Color	-	Green	Green
Design Permissible Shear Partially Vegetated ²	Design Value	7.0 lb/ft ²	336 PA
Design Permissible Shear Fully Vegetated ²	Design Value	12.0 lb/ft ²	576 PA
Design Velocity Vegetated ²	Design Value	20 ft/sec	6.1 m/sec
Mannings n ³	Design Value	0.026 - 0.036	0.026 - 0.036
Roll Dimensions	-	8.5 x 90 ft	2.6 x 27.4 m
Roll Area	-	765 ft ²	72.07 m ²

*Minimum Average Roll Value (MARV)

²Maximum permissible velocity and shear stress have been obtained utilizing a rigorous vegetated testing program with specific soil types, vegetation retardance classes, flow conditions, and failure criteria. These conditions may not be relevant to every project or other site conditions.

³Manning's n will vary depending on flow depth.

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